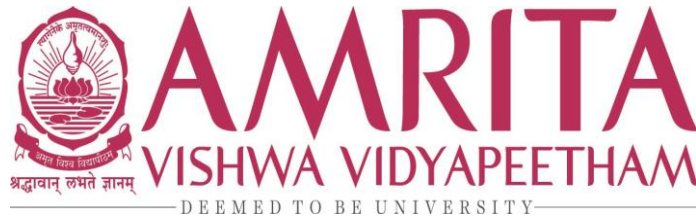


**AMRITA SCHOOL OF ENGINEERING, COIMBATORE**

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## IEEE 9-Bus System

### Objective

To simulate the IEEE 9-bus system using PSS®E and carry-out its power flow and short-circuit analysis

### Introduction

This lab is designed to be an open-ended lab unlike the previous labs where your response was mostly guided. This lab allows you to work on the *simulation based project* having weightage of 14% of the total lab grade. The project is related to modelling of standard IEEE 9-bus system followed by its power-flow and short-circuit analysis. The task wise implementation is listed in the following section. You are expected to implement and practice the skills learnt and the knowledge gained in the previous lab sessions and theory classes. The maximum number of students working on the project together is 3.

### Deliverables

The task wise details of the project are listed below.

#### Task 1: To obtain the IEEE 9-bus system data

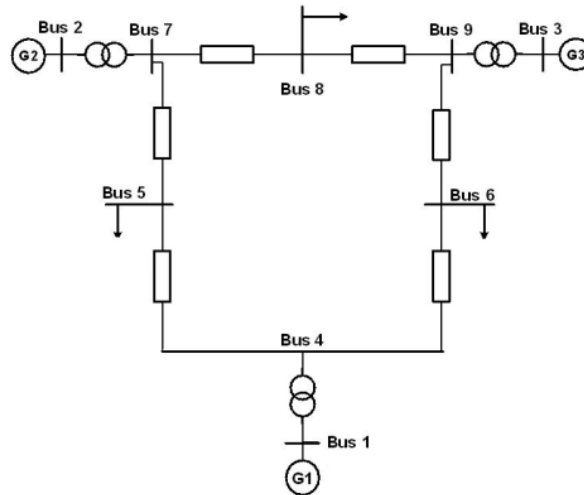
##### 1. Discuss what do you understand by IEEE and its standard transmission test bus-systems.

Standard test systems for transmission networks are used to model real-world networks where researchers try to implement new ideas and concepts. Transmission systems are usually balanced in design along with heavy presence of generation. Several test systems exist for separately studying Transmission networks (TNs). For TN studies, the most widely used systems are the ones developed by IEEE, for power flow, and transient stability studies. The Institute of Electrical and Electronics Engineers (IEEE) is a professional association for electrical and electronics engineering which is dedicated to advancing technology for the benefit of humanity. As of 2018, it is the world's largest association of technical professionals with more than 423,000 members in over 160 countries around the world. Some of the IEEE standard transmission test bus-systems are the 9, 14, 30, 39 and 118 bus systems. Other research groups and organizations have also developed their own test systems; for example, the 150- bus synthetic system from the University of Illinois, UrbanaChampaign, the 9-bus and 179-bus systems from the Western System Coordinating Council, used for transient stability studies.

2. Obtain the 9-bus system data using online resources (preferably research papers) required to simulate the system. This includes the following:

- Network layout and operating voltages
- Line data (impedance / admittance)
- Load data (load MW/MVAR)
- Generator data (type, impedance, ratings etc.)
- Transformer data (impedance, ratings etc.)

Network layout:



Bus Operating Voltages:

| Bus | Type              | Voltage (kV) | Voltage (pu) |
|-----|-------------------|--------------|--------------|
| 1   | Swing Bus         | 16.5         | 1.04         |
| 2   | Generator Bus     | 18           | 1.025        |
| 3   | Generator Bus     | 13.8         | 1.025        |
| 4   | Non-Generator Bus | 230          | 1.000        |
| 5   | Non-Generator Bus | 230          | 1.000        |
| 6   | Non-Generator Bus | 230          | 1.000        |
| 7   | Non-Generator Bus | 230          | 1.000        |

|   |                   |     |       |
|---|-------------------|-----|-------|
| 8 | Non-Generator Bus | 230 | 1.000 |
| 9 | Non-Generator Bus | 230 | 1.000 |

**Transmission line data:**

| Line     |        | R [pu] | X [pu] | B [pu] |
|----------|--------|--------|--------|--------|
| From Bus | To Bus |        |        |        |
| 4        | 5      | 0.0100 | 0.0850 | 0.1760 |
| 4        | 6      | 0.0170 | 0.0920 | 0.1580 |
| 5        | 7      | 0.0320 | 0.1610 | 0.3060 |
| 6        | 9      | 0.0390 | 0.1700 | 0.3580 |
| 7        | 8      | 0.0085 | 0.0720 | 0.1490 |
| 8        | 9      | 0.0119 | 0.1008 | 0.2090 |

**Load Data:**

| Bus | P (MW) | Q (MVAR) |
|-----|--------|----------|
| 5   | 125    | 50       |
| 6   | 90     | 30       |
| 8   | 100    | 35       |

**Transformer Data:**

| Transformer |        | R [pu] | X [pu] |
|-------------|--------|--------|--------|
| From Bus    | To Bus |        |        |
| 2           | 7      | 0.000  | 0.0625 |
| 9           | 3      | 0.000  | 0.0586 |
| 4           | 1      | 0.000  | 0.0576 |

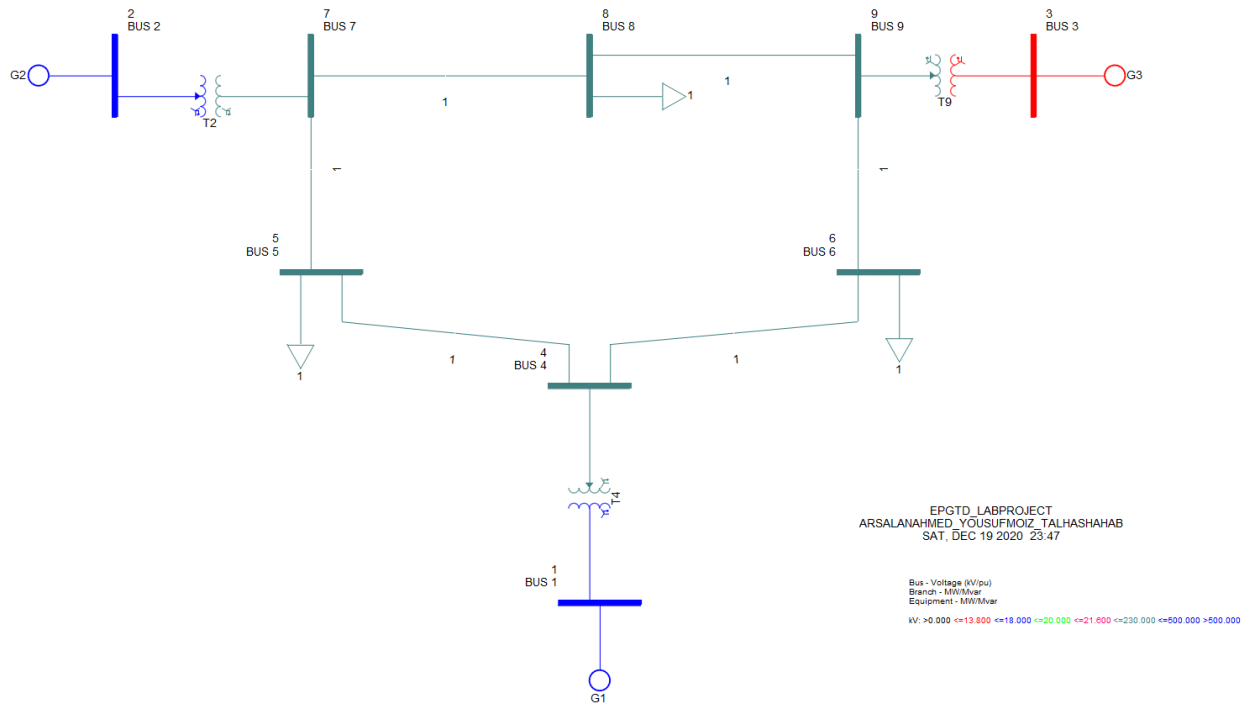
**Generator Data:**

| Bus | Type          | V [pu] | Pgen (MW) | R [pu] | X [pu] |
|-----|---------------|--------|-----------|--------|--------|
| 1   | Swing         | 1.04   | -         | 0.0    | 0.2    |
| 2   | Generator Bus | 1.025  | 163       | 0.0    | 0.15   |
| 3   | Generator Bus | 1.025  | 85        | 0.0    | 0.25   |

## Task 2: To simulate the IEEE 9-bus system using PSS®E

1. Using the obtained data, create a single-line diagram on PSS®E with its associated network data that represents the standard 9-bus system. Each component used in the single-line diagram should be named properly. Add meaningful legend and title to your single-line diagram file using the options available in the software tool.

### Single-line diagram:



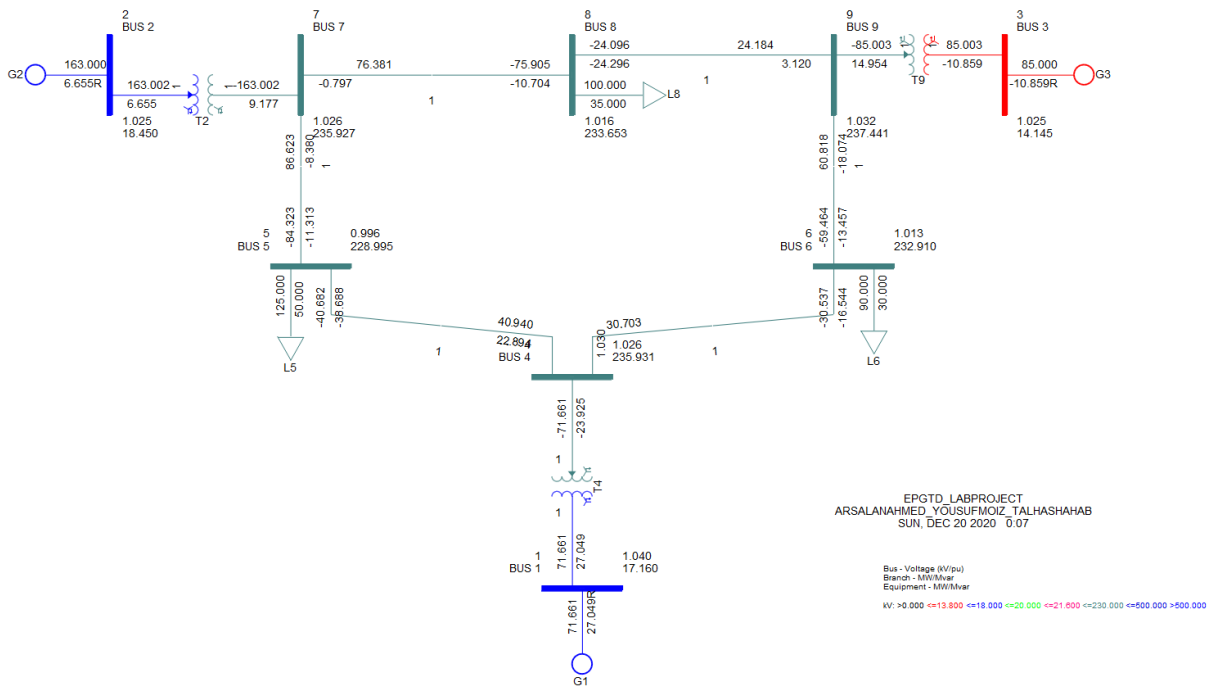
### 2. Identify the layout of this transmission network.

The layout of the transmission network shown above can be considered to be an **interconnected system**. This is because the feeder ring is being energized by multiple generating stations.

### Task 3: To obtain the power-flow solution

- Obtain the power-flow solution of the system. Generate its report and display the results on single-line diagram.
  - Active (MW) and Reactive (MVAR) Power with arrows to indicate the direction
  - Bus Voltage (per-unit and kV)

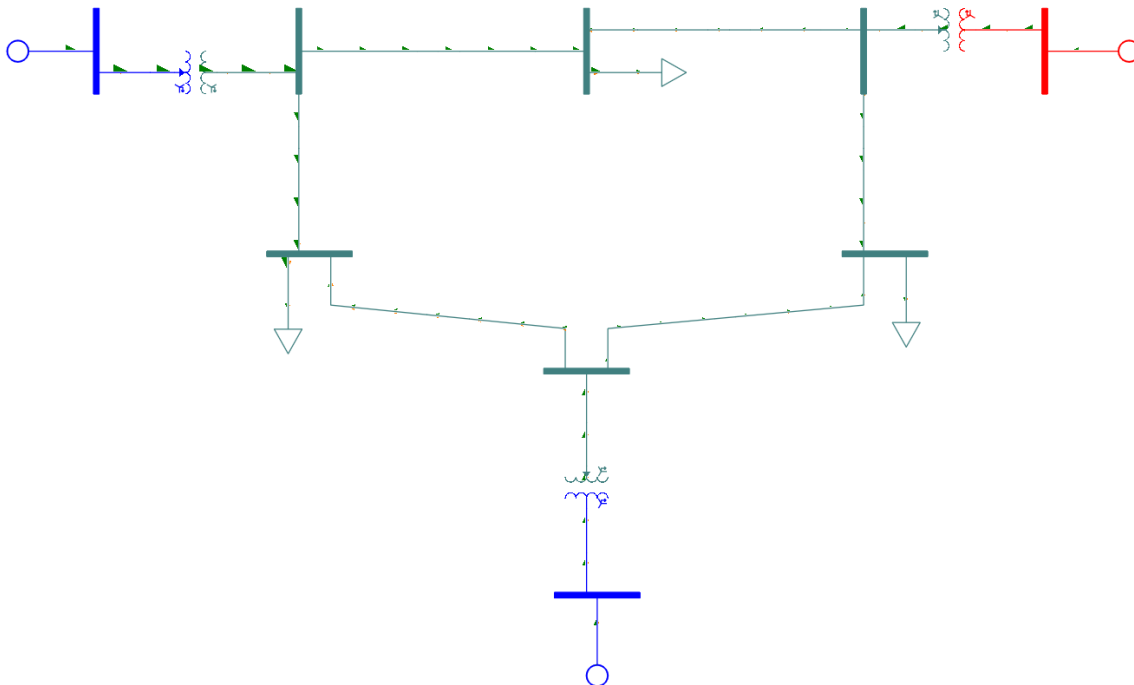
#### Load Flow Analysis with bus voltages:



**Load Flow report:**

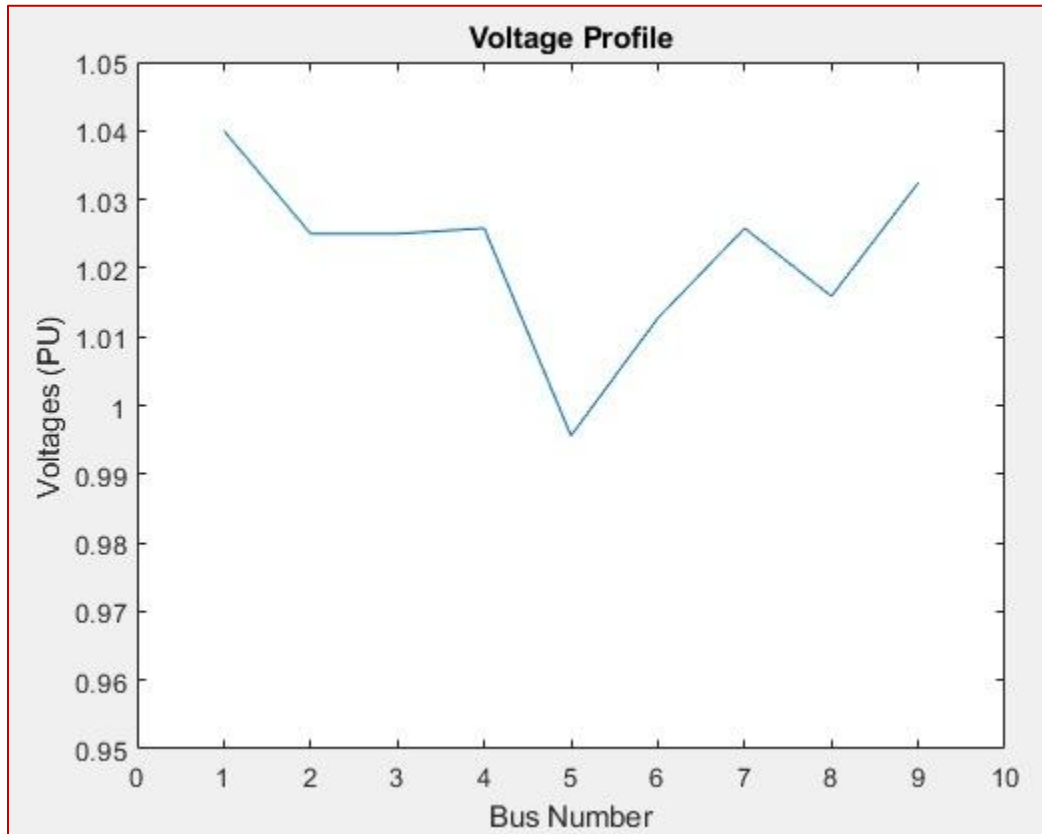
| PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS(R)E |   |       |        |     |        |        |       |     |          | SUN, DEC 20 2020 |      | 0:16                             |       |       |      |        |       |      |        |   |
|-------------------------------------------------|---|-------|--------|-----|--------|--------|-------|-----|----------|------------------|------|----------------------------------|-------|-------|------|--------|-------|------|--------|---|
| EPGTD LABPROJECT                                |   |       |        |     |        |        |       |     |          | RATING           |      | %MVA FOR TRANSFORMERS            |       |       |      |        |       |      |        |   |
| ARSALANAHMED_YOUSUFMOIZ_TALHASHAHAB             |   |       |        |     |        |        |       |     |          | SET 1            |      | % I FOR NON-TRANSFORMER BRANCHES |       |       |      |        |       |      |        |   |
| BUS                                             | 1 | BUS 1 | 16.500 | CKT | MW     | MVAR   | MVA   | %   | 1.0400PU | 0.00             | X--- | LOSSES                           | ---X  | X---- | AREA | -----X | X---- | ZONE | -----X | 1 |
| FROM GENERATION                                 |   |       |        |     | 71.7   | 27.0R  | 76.6  | 77  | 17.160KV |                  |      | MW                               | MVAR  | 1     |      |        |       | 1    |        |   |
| TO                                              | 4 | BUS 4 | 230.00 | T4  | 71.7   | 27.0   | 76.6  |     | 1.0000UN |                  |      | 0.00                             | 3.12  | 1     |      |        |       | 1    |        |   |
| BUS                                             | 2 | BUS 2 | 18.000 | CKT | MW     | MVAR   | MVA   | %   | 1.0250PU | 9.28             | X--- | LOSSES                           | ---X  | X---- | AREA | -----X | X---- | ZONE | -----X | 2 |
| FROM GENERATION                                 |   |       |        |     | 163.0  | 6.7R   | 163.1 | 163 | 18.450KV |                  |      | MW                               | MVAR  | 1     |      |        |       | 1    |        |   |
| TO                                              | 7 | BUS 7 | 230.00 | T2  | 163.0  | 6.7    | 163.1 |     | 1.0000LK |                  |      | 0.00                             | 15.83 | 1     |      |        |       | 1    |        |   |
| BUS                                             | 3 | BUS 3 | 13.800 | CKT | MW     | MVAR   | MVA   | %   | 1.0250PU | 4.66             | X--- | LOSSES                           | ---X  | X---- | AREA | -----X | X---- | ZONE | -----X | 3 |
| FROM GENERATION                                 |   |       |        |     | 85.0   | -10.9R | 85.7  | 86  | 14.145KV |                  |      | MW                               | MVAR  | 1     |      |        |       | 1    |        |   |
| TO                                              | 9 | BUS 9 | 230.00 | T9  | 85.0   | -10.9  | 85.7  |     | 1.0000UN |                  |      | 0.00                             | 4.10  | 1     |      |        |       | 1    |        |   |
| BUS                                             | 4 | BUS 4 | 230.00 | CKT | MW     | MVAR   | MVA   | %   | 1.0258PU | -2.22            | X--- | LOSSES                           | ---X  | X---- | AREA | -----X | X---- | ZONE | -----X | 4 |
| TO                                              | 1 | BUS 1 | 16.500 | T4  | -71.7  | -23.9  | 75.5  |     | 1.0000LK |                  |      | MW                               | MVAR  | 1     |      |        |       | 1    |        |   |
| TO                                              | 5 | BUS 5 | 230.00 | 1   | 40.9   | 22.9   | 46.9  |     |          |                  |      | 0.26                             | 2.19  | 1     |      |        |       | 1    |        |   |
| TO                                              | 6 | BUS 6 | 230.00 | 1   | 30.7   | 1.0    | 30.7  |     |          |                  |      | 0.17                             | 0.90  | 1     |      |        |       | 1    |        |   |
| BUS                                             | 5 | BUS 5 | 230.00 | CKT | MW     | MVAR   | MVA   | %   | 0.9956PU | -3.99            | X--- | LOSSES                           | ---X  | X---- | AREA | -----X | X---- | ZONE | -----X | 5 |
| TO LOAD-PQ                                      |   |       |        |     | 125.0  | 50.0   | 134.6 |     |          |                  |      | MW                               | MVAR  | 1     |      |        |       | 1    |        |   |
| TO                                              | 4 | BUS 4 | 230.00 | 1   | -40.7  | -38.7  | 56.1  |     |          |                  |      | 0.26                             | 2.19  | 1     |      |        |       | 1    |        |   |
| TO                                              | 7 | BUS 7 | 230.00 | 1   | -84.3  | -11.3  | 85.1  |     |          |                  |      | 2.30                             | 11.57 | 1     |      |        |       | 1    |        |   |
| BUS                                             | 6 | BUS 6 | 230.00 | CKT | MW     | MVAR   | MVA   | %   | 1.0127PU | -3.69            | X--- | LOSSES                           | ---X  | X---- | AREA | -----X | X---- | ZONE | -----X | 6 |
| TO LOAD-PQ                                      |   |       |        |     | 90.0   | 30.0   | 94.9  |     |          |                  |      | MW                               | MVAR  | 1     |      |        |       | 1    |        |   |
| TO                                              | 4 | BUS 4 | 230.00 | 1   | -30.5  | -16.5  | 34.7  |     |          |                  |      | 0.17                             | 0.90  | 1     |      |        |       | 1    |        |   |
| TO                                              | 9 | BUS 9 | 230.00 | 1   | -59.5  | -13.5  | 61.0  |     |          |                  |      | 1.35                             | 5.90  | 1     |      |        |       | 1    |        |   |
| BUS                                             | 7 | BUS 7 | 230.00 | CKT | MW     | MVAR   | MVA   | %   | 1.0258PU | 3.72             | X--- | LOSSES                           | ---X  | X---- | AREA | -----X | X---- | ZONE | -----X | 7 |
| TO                                              | 2 | BUS 2 | 18.000 | T2  | -163.0 | 9.2    | 163.3 |     | 1.0000UN |                  |      | MW                               | MVAR  | 1     |      |        |       | 1    |        |   |
| TO                                              | 5 | BUS 5 | 230.00 | 1   | 86.6   | -8.4   | 87.0  |     |          |                  |      | 0.00                             | 15.83 | 1     |      |        |       | 1    |        |   |
| TO                                              | 8 | BUS 8 | 230.00 | 1   |        |        |       |     |          |                  |      |                                  |       |       |      |        |       |      |        |   |

**Load Flow analysis indicating flow of power:**



2. Using the results of power flow analysis, plot a ***voltage profile*** (bus voltages in pu on y-axis and number of bus at x-axis).

**Voltage Profile:**



The above graph can be used to find out the respective per unit voltage for each bus in the network.

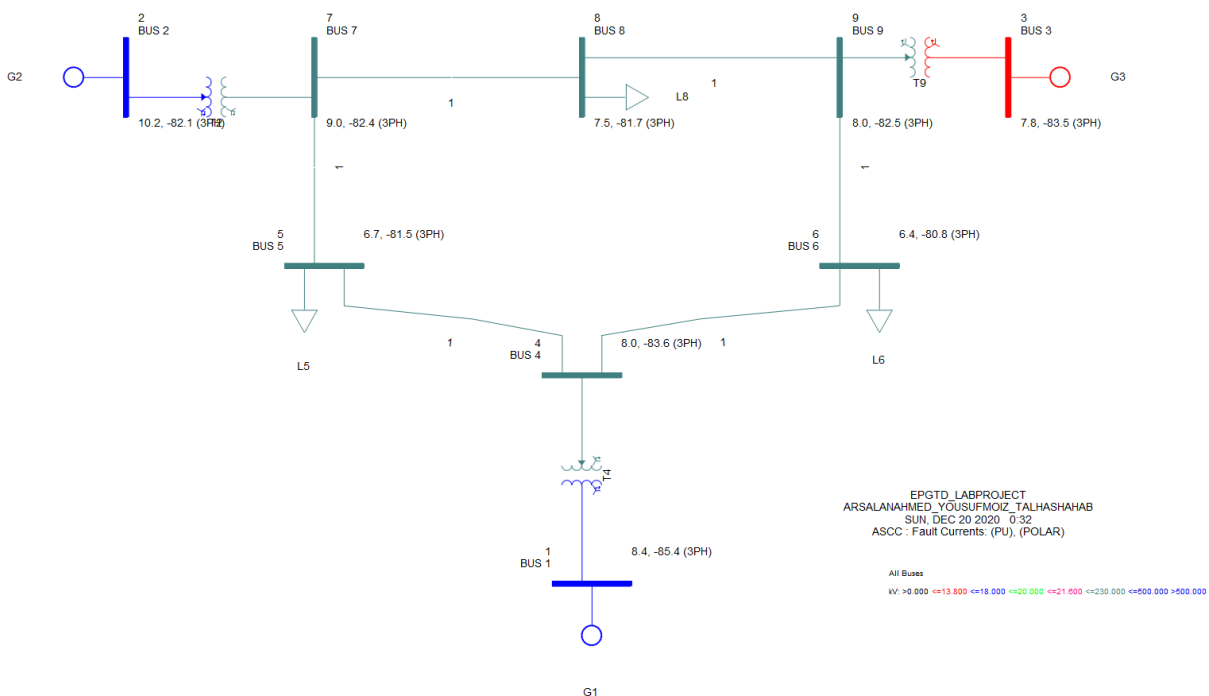
## Task 4: To perform short-circuit analysis

1. Carry-out short-circuit analysis for three-phase bolted faults on all buses.

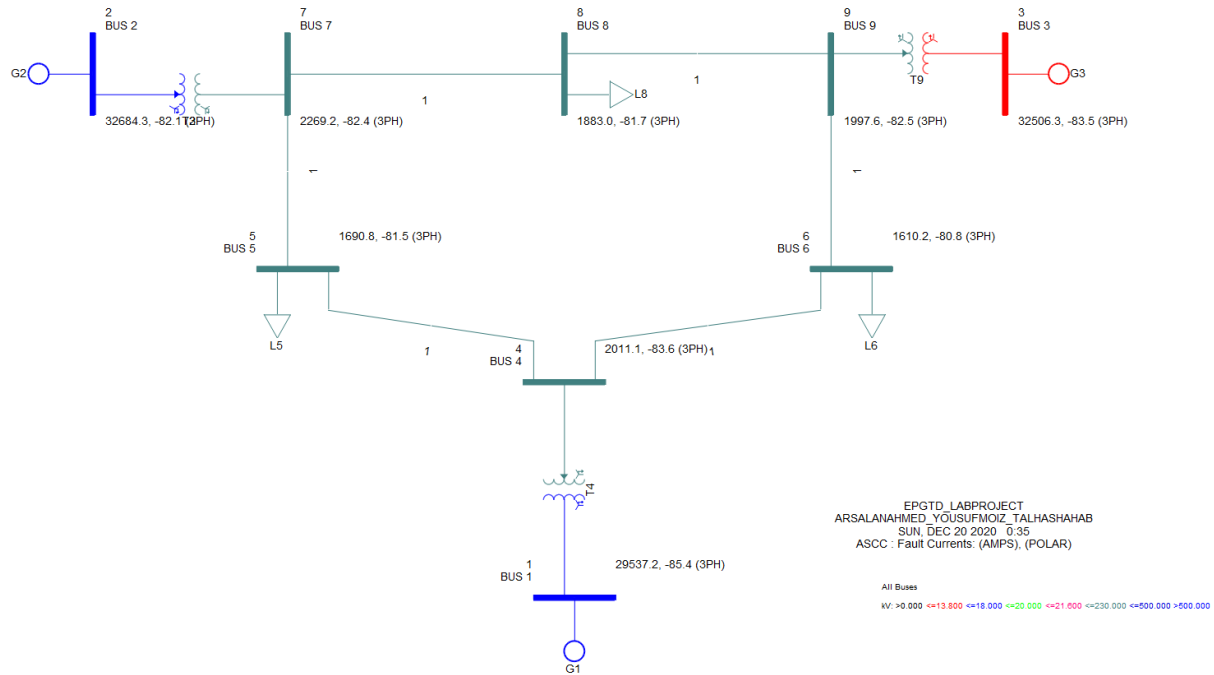
Since we were simulating balanced faults, we did not need to give the values of sequence impedances, however, we required the values of resistance and reactance of the generators. The resistance and reactance data that was used for the short circuit analysis can be found in the generator data table in task 1.

2. Display the results on single-line diagram (Current in Amperes in Polar form).

### Short Circuit Analysis (values in PU):



**Short Circuit Analysis (values in Amperes):**



3. Generate the report.

### Short Circuit analysis report (values in PU):

PSS(R)E-34.3.2 ASCC SHORT CIRCUIT CURRENTS                      SUN, DEC 20 2020    0:39

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OPTIONS USED:

- ```
- SET PRE-FAULT VOLTAGES AND PHASE SHIFT ANGLES TO POWER FLOW SOLUTION
- SET SYNCHRONOUS/ASYNCHRONOUS MACHINE POWER OUTPUTS TO P=0.0, Q=0.0
- SET GENERATOR POSITIVE SEQUENCE REACTANCES TO ZSORCE
- SET INDUCTION MACHINE POSITIVE SEQUENCE REACTANCES TO SUBTRANSIENT
- TRANSFORMER TAP RATIOS AND PHASE SHIFT ANGLES UNCHANGED
- SET LINE CHARGING=0.0 IN +/-0 SEQUENCES
- SET LINE/FIXED/SWITCHED SHUNTS=0.0 AND TRANSFORMER MAGNETIZING ADMITTANCE=0.0 IN +/-0 SEQUENCES
- SET LOAD=0.0 IN +/-0 SEQUENCES
- DC LINES AND FACTS DEVICES BLOCKED
- IMPEDANCE CORRECTIONS NOT APPLIED TO TRANSFORMER ZERO SEQUENCE IMPEDANCES
```

| X-----BUS-----X |        |         |    | THREE PHASE FAULT |             |
|-----------------|--------|---------|----|-------------------|-------------|
|                 |        |         |    | /I+/<br>PU        | AN(I+)<br>- |
| 1               | [BUS 1 | 16.500] | PU | 8.4414            | -85.45      |
| 2               | [BUS 2 | 18.000] | PU | 10.1899           | -82.12      |
| 3               | [BUS 3 | 13.800] | PU | 7.7698            | -83.49      |
| 4               | [BUS 4 | 230.00] | PU | 8.0115            | -83.62      |
| 5               | [BUS 5 | 230.00] | PU | 6.7356            | -81.51      |
| 6               | [BUS 6 | 230.00] | PU | 6.4144            | -80.82      |
| 7               | [BUS 7 | 230.00] | PU | 9.0400            | -82.37      |
| 8               | [BUS 8 | 230.00] | PU | 7.5012            | -81.74      |
| 9               | [BUS 9 | 230.00] | PU | 7.9579            | -82.53      |

## Short Circuit analysis report (values in Amperes):

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ARSALANAHMED\_YOUSUFMOIZ\_TALHASHAHAB

PSS(R)E-34.3.2 ASSC SHORT CIRCUIT CURRENTS

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### OPTIONS USED:

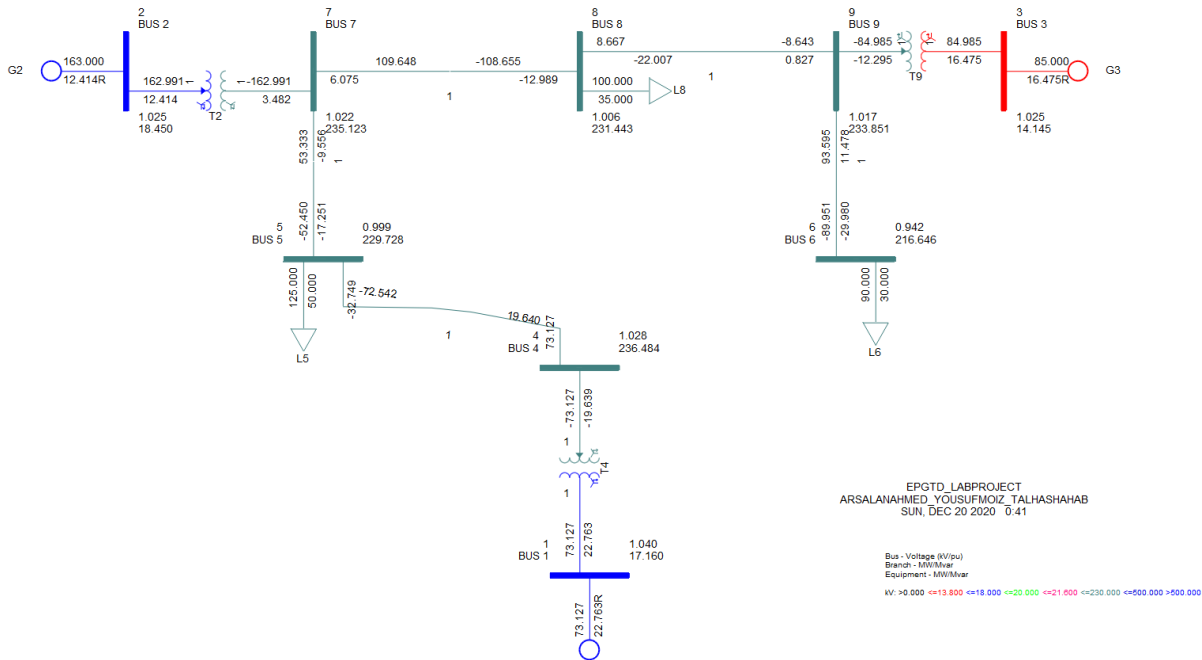
- SET PRE-FAULT VOLTAGES AND PHASE SHIFT ANGLES TO POWER FLOW SOLUTION
- SET SYNCHRONOUS/ASYNCHRONOUS MACHINE POWER OUTPUTS TO P=0.0, Q=0.0
- SET GENERATOR POSITIVE SEQUENCE REACTANCES TO ZSORCE
- SET INDUCTION MACHINE POSITIVE SEQUENCE REACTANCES TO SUBTRANSIENT
- TRANSFORMER TAP RATIOS AND PHASE SHIFT ANGLES UNCHANGED
- SET LINE CHARGING=0.0 IN +/-0 SEQUENCES
- SET LINE/FIXED/SWITCHED SHUNTS=0.0 AND TRANSFORMER MAGNETIZING ADMITTANCE=0.0 IN +/-0 SEQUENCES
- SET LOAD=0.0 IN +/-0 SEQUENCES
- DC LINES AND FACTS DEVICES BLOCKED
- IMPEDANCE CORRECTIONS NOT APPLIED TO TRANSFORMER ZERO SEQUENCE IMPEDANCES

| X-----X |        |         | THREE PHASE FAULT |        |
|---------|--------|---------|-------------------|--------|
|         | BUS    |         | /I+/<br>AMP       | AN(I+) |
| 1       | [BUS 1 | 16.500] | 29537.2           | -85.45 |
| 2       | [BUS 2 | 18.000] | 32684.3           | -82.12 |
| 3       | [BUS 3 | 13.800] | 32506.3           | -83.49 |
| 4       | [BUS 4 | 230.00] | 2011.1            | -83.62 |
| 5       | [BUS 5 | 230.00] | 1690.8            | -81.51 |
| 6       | [BUS 6 | 230.00] | 1610.2            | -80.82 |
| 7       | [BUS 7 | 230.00] | 2269.2            | -82.37 |
| 8       | [BUS 8 | 230.00] | 1883.0            | -81.74 |
| 9       | [BUS 9 | 230.00] | 1997.6            | -82.53 |

## Task 5: To modify the network and apply techniques of voltage regulation

- Now modify the network such that the line connecting bus 4 and bus 6 is removed. Re-run the power-flow and obtain bus voltages.

### Power Flow results after removing bus 4 to bus 6 line:



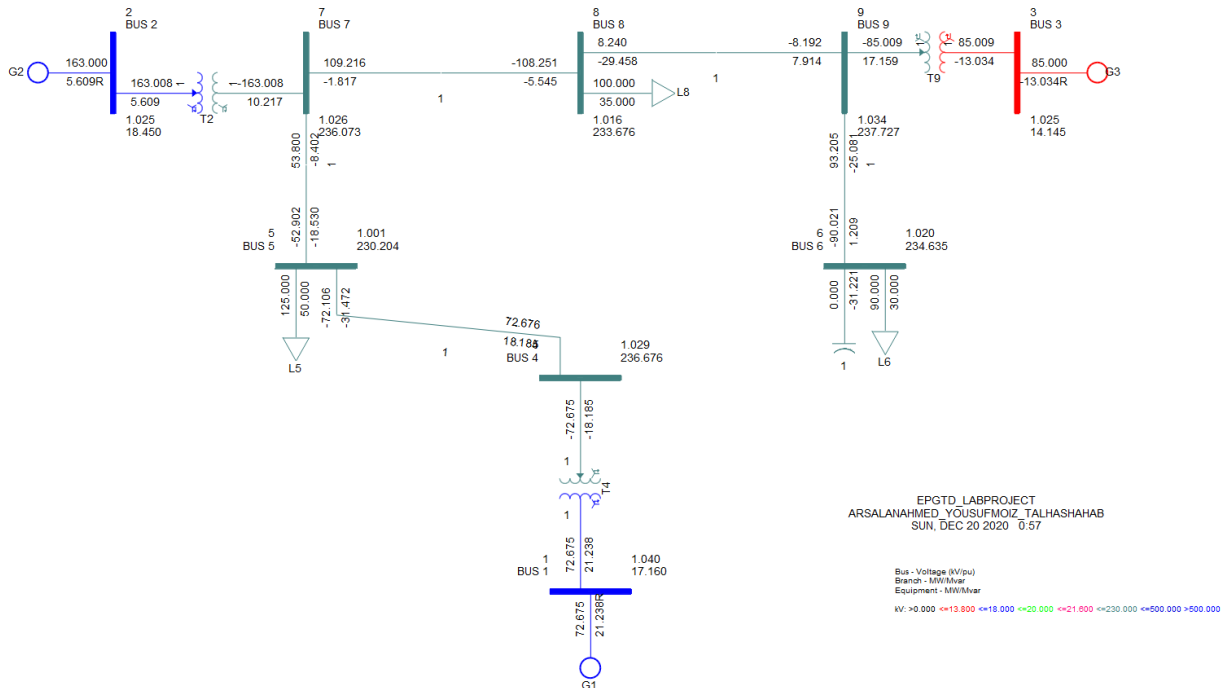
- What is the impact of this modification on the supply to the loads? Is supply to any of the loads interrupted?

When the connection between the bus 4 and bus 6 was interrupted, most of the buses were not affected which shows the advantages of using an interconnected system. However, the load connected at bus 6 was affected as the voltage of the bus reduced from 1.013 pu to 0.942 pu which would have affected the supply to the load connected.

- If the resulting voltage at any bus has been decreased below 0.96 pu, apply technique(s) to improve the voltage regulation.

**Technique used:** Adding a shunt capacitor supplying reactive power to the load.

## Updated power flow results after addition of shunt capacitor:



In the above single-line diagram, the shunt capacitor added has a rating of 30 MVAR, which is equal to the rating of the load in parallel. After running the load flow on the above network, it can be seen that voltage regulation of the bus 6 has improved and the voltage has increased from 0.942 pu to 1.020 pu.

## **Task 6: To identify the protection schemes required for protection of different components of the system**

1. There are multiple components in the system (generators, transmission lines, transformers, feeders /loads, bus bars). Identify few protection schemes that can be applied to protect these components/systems in case of any potential fault. Discuss your recommendations and justify them with reasons.

The following protection schemes can be used to protect these components:

- **Over-Current Relays:**

The major function of an over-current relay is to detect whether the current flowing in the line has exceeded a set threshold. These relays can be used to protect the system from current surges by generating trip signals for the circuit breakers, thus breaking supply to the line. Definite time over-current relay (DTOCR) is also an over-current relay but with the added functionality of generating a trip signal only after waiting a definite amount of time. Therefore, if the fault is only minor and disappears after a certain time, the relay would not generate the trip signal and the supply to the line would not be interrupted.

- **Differential Relay:**

Differential relays work on the principle that the current entering a component should be the same as the current leaving the component. Therefore, they constantly monitor the current entering and leaving a component and whenever they sense any difference between the two currents, a trip signal is generated for the circuit breakers, thus breaking supply to the line. Hence, differential relays can be beneficial for the protection of equipment such as transformers, generators and loads.

- **Primary and Backup Protection:**

We can incorporate primary and backup protection in our system to provide two levels of protection. In the primary protection, we can protect each of the individual lines with an overcurrent relay, where a fault in a specific line would be cleared by the relay and circuit breaker connected to that line. In cases where for some reason, the primary relays are not able to clear the fault, the backup protection can takeover. In the backup protection, the over-current relays will be employed at the generating stations with sufficient time delay which first provides an opportunity for the primary relays to take action, and only takes over if the primary protection fails to take action in that time and cuts off supply to the whole transmission line.

**Task 7: To write a report concluding your work**

1. Write a report to summarize all the work done in the above tasks. The report should contain task wise results and analysis where required.
2. The data and results should be presented in the form of tables. Include the single-line diagram representation of the obtained results. Attach the reports generated by the software in Appendices. All the tables and figures must be labeled in a meaningful way.